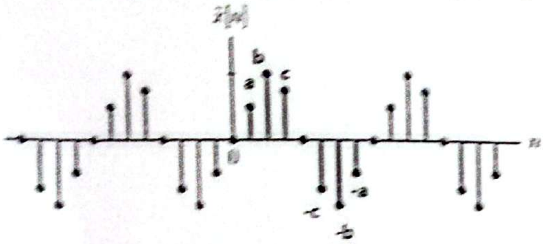
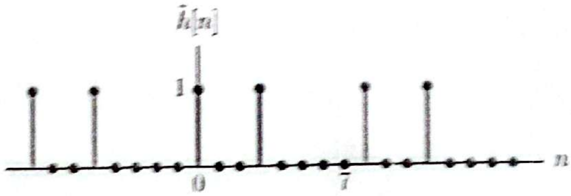
Habib University
Course Code: EE252
Course Title: Signals and Systems
Instructor's Name: Tariq Mumtaz
Examination: Quiz 3 Date: 24th April, 2025
Total Marks: 50 (5% Course credit) Duration:50 minutes

CLO: 03; CDL: 3, PLO: 1

Consider the two periodic discrete signals shown below. Take any random values for a, b, and c in the range of 1 to 10. Make sure (b > c > a).





I. [10] Compute the output y [n] using <u>periodic convolution</u>, show all steps of working, including plots of h[n–k] and an accurate expression of y[n] for values of **n = 0 to 3**

$$\tilde{y}[n] = \tilde{x}[n] \otimes \tilde{h}[n] = \sum_{k=0}^{N-1} \tilde{x}[k]\,\tilde{h}[n-k]\,, \quad \text{all } n$$

II. [10] Compute the output y [n] using <u>circular convolution</u>, show all steps of working, including arrays of x[n] and h[n–k] for values of **n = 4 to 7**

III. [10] Write the accurate <u>expression</u> of DTFS coefficients (C_k) of signal y[n] (as calculated in part- I) by showing the calculation for angular frequency (Ω_K) and identifying the number of required basis functions (φ_k).

Synthesis equation:

$$\tilde{x}[n] = \sum_{k=0}^{N-1} \tilde{c}_k\, e^{j(2\pi/N)kn}\,, \qquad \text{all } n$$

Analysis equation:

$$\tilde{c}_k = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}[n]\, e^{-j(2\pi/N)kn}$$

IV. [20] Using MATLAB, determine the <u>value</u> of the DTFS coefficients of signals x[n], h[n], and y[n]. Verify that the periodic convolution property holds. For submission, provide the code to calculate DTFS, the DTFS coefficient values, and the steps involved in the proof.

$$\tilde{x}[n] \overset{\text{DTFS}}{\longleftrightarrow} \tilde{c}_k \qquad \text{and} \qquad \tilde{h}[n] \overset{\text{DTFS}}{\longleftrightarrow} \tilde{d}_k$$

$$\tilde{x}[n] \otimes \tilde{h}[n] \overset{\text{DTFS}}{\longleftrightarrow} N\,\tilde{c}_k\,\tilde{d}_k$$

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I

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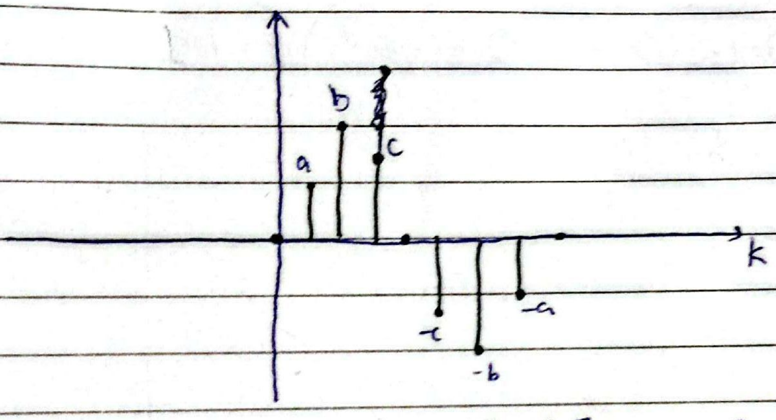
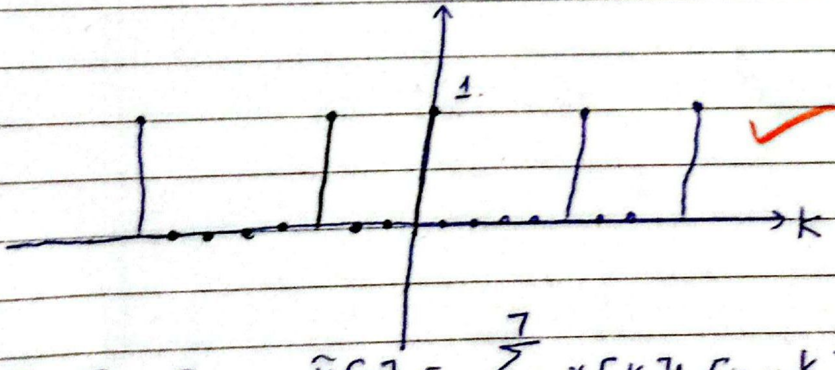
Q.) $x[n] = \{0, a, b, c, 0, -c, -b, -a\}$
$h[n] = \{1, 0, 0, 1, 0, 0, 0, 0\}$

I) If $n = 0$:

$\tilde{x}[k]$



$h[n-k] \approx n = 0$



$$\tilde{y}[n] = \sum_{k=0}^{7} x[k]\,h[n-k]$$

h[-1]
h[-2]

$$\tilde{y}[0] = x[0]h[0] + x[1]h[1] + x[2]h[2] + x[3]h[3] + x[4]h[4] + x[5]h[5] + x[6]h[6] + x[7]h[7]$$

$$= 0\cdot 1 + a\cdot 0 + b\cdot 0 + c\cdot 0 + 0\cdot 0 + (-c)(1) + (-b)(0) + (-a) = 0$$

$$\boxed{y[0] = -c}$$

II

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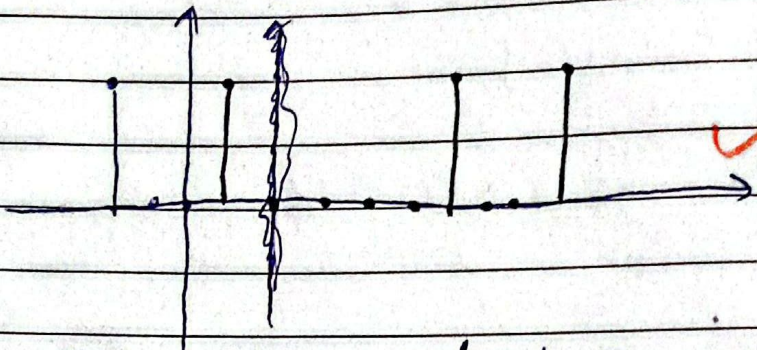
If $n=1$:

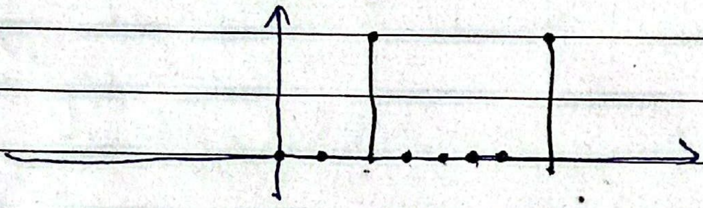
$h[1-k]$



$$y[1] = 0\cdot 0 + a\cdot 1 + b\cdot 0 + c\cdot 0 + 0\cdot 1 + (-c\cdot 0) + (-b\cdot 1) + (-a)\cdot 0.$$

$$\boxed{y[1] = a - b}$$

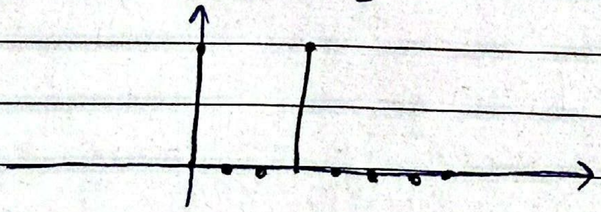
If $n=2$: $h[2-k]$



$$y[2] = 0\cdot 0 + a\cdot 0 + b\cdot 1 + c\cdot 0 + 0\cdot 0 + (-c\cdot 0) + (-b\cdot 0) + (-a)\cdot 1$$

$$\boxed{y[2] = b - a}$$

If $n=3$: $h[3-k]$



$$y[3] = 0\cdot 1 + a\cdot 0 + b\cdot 0 + c\cdot 1 + 0\cdot 0 + (-c\cdot 0) + (-b\cdot 0) + (-a\cdot 0).$$

$$\boxed{y[3] = c}$$

$$\boxed{y[n] = \{-c, \underset{\substack{\uparrow \\ n=0}}{a-b}, b-a, c\}}$$

01000010 $n=1$
00100001 $n=2$
10010000 $n=3$

III

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II) $x[n] = \{0, a, b, c, 0, -c, -b, -a\}$.
$h[n] = \{1, 0, 0, 1, 0, 0, 0, 0\}$.

+10

$n=4$:

$x\{k\} = \{0, a, b, c, 0, -c, -b, -a\}$.
$h[n-k] = \{0, 1, 0, 0, 1, 0, 0, 0\}$.

$\tilde{y}[4] = 0\cdot 0 + a\cdot 1 + b\cdot 0 + c\cdot 0 + 0\cdot 1 + (-c\cdot 0) + (-b\cdot 0) + (-a\cdot 0)$

$\boxed{y[4] = a}$.

$n=5$:

$h[5-k] = \{0, 0, 1, 0, 0, 1, 0, 0\}$.

$y[5] = (0\cdot 0) + (a\cdot 0) + (b\cdot 1) + (c\cdot 0) + (0\cdot 0) + (-c\cdot 1) + (-b\cdot 0) + (-a\cdot 0)$

$\boxed{y[5] = b - c}$.

$n=6$:

$h[6-k] = \{0, 0, 0, 1, 0, 0, 1, 0\}$.

$y[6] = (0\cdot 0) + (a\cdot 0) + (b\cdot 0) + (c\cdot 1) + (0\cdot 0) + (-c\cdot 0) + (-b\cdot 1) + (-a\cdot 0)$

$\boxed{y[6] = c - b}$

$n=7$:

$h[7-k] = \{0, 0, 0, 0, 1, 0, 0, 1\}$.

$y[7] = (0\cdot 0) + (a\cdot 0) + (b\cdot 0) + (c\cdot 0) + (0\cdot 1) + (-c\cdot 0) + (-b\cdot 0) + (-a\cdot 1)$

$\boxed{y[7] = -a}$.

IV

$y[n] = \{\underset{\substack{\uparrow \\ n=4}}{a}, b-c, c-b, -a\}$

$y[n] = \{\underset{\substack{\uparrow \\ n=0}}{-c}, a-b, b-a, c, a, b-c, c-b, -a\}$

III) $y[n] = \{-c, a-b, b-a, c, a, b-c, c-b, -a\}$.

$\Omega = \frac{2\pi}{N}$

$$Y_k = \frac{1}{8}\sum_{k=0}^{8-1} y[n]\, e^{-j\left(\frac{2\pi}{8}\right)kn}$$

→ Number of required basis functions is 8.

$$Y_k = \frac{1}{8}(-c)\cancel{e^{-j\pi/4 k(0)}}^{\,1} + \frac{a-b}{8}e^{-j\pi/4 k(1)} + \frac{b-a}{8}e^{-j\frac{\pi}{4}k(2)} + \frac{c}{8}e^{-j\pi/4 k(3)}$$
$$+ \frac{a}{8}e^{-j\pi/4 k(4)} + \frac{b-c}{8}e^{-j\pi/4 k(5)} + \frac{c-b}{8}e^{-j\pi/4 k(6)} + \frac{(-a)}{8}e^{-j\pi/4 k(7)}.$$

$$\boxed{Y_k = \frac{1}{8}\left[-c + (a-b)e^{-j\pi/4 k} + (b-a)e^{-j\frac{\pi}{2}k} + (c)e^{-j\,3\pi/4} + a e^{-j\pi k} + (b-c)e^{-j\,5\pi/4 k} + (c-b)e^{-j\frac{3\pi}{2}k} + (-a)e^{-j\frac{7\pi}{4}k}\right]}$$

Forgot to assume values for a, b, and c.

* Assuming $a = 1$, $b = 3$, $c = 2$

I) $y[n] = \{\underset{\substack{\uparrow \\ n=0}}{-2}, -2, 2, 2\}$

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(V)

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II) $y[n] = \{\underset{\underset{n=4}{\uparrow}}{1}, 1, -1, -1\}$

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$y[n] = \{-2, -2, 2, 2, 1, \underline{1}, -1, -1\}.$

III) $\psi_k = \frac{1}{8}\Big[-2 + -2e^{-j\pi/4 k} + 2e^{-j\frac{\pi}{2}k} + 2e^{-j\frac{3\pi}{4}} + e^{-j\pi k} + e^{-j\frac{5\pi}{4}k}$
$- e^{-j\,3\pi/2\,k} - e^{-j\frac{7\pi}{4}k}\Big].$

+10

IV)

$c_0 = 0.0 + j0.0 = 0$
$c_1 = 0 - j1.2803 = -1.2803j$
$c_2 = 0 + j0.25 = 0.25j$
$c_3 = 0 + j0.2197 = 0.2197j$
$c_4 = 0 + 0i = 0$

c_k

Should be 8 values

$d_0 = 0.25$
$d_1 = 0.0366 - 0.0884j$
$d_2 = 0.1250 + 0.1250j$
$d_3 = 0.2134 - 0.0884j$
$d_4 = 0.$

d_k.

$\cancel{y_{16} = -1 + 0.25j}$

MATLAB

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$Y_0 = -1 + 0.25j$

$Y_1 = 0.2269 - 1.7208j$

$Y_2 = 0.8047 + 0.3499j$

$Y_3 = 0.6016 + 0.6432j$

$Y_4 = -0.6333 + 0.4778j$

$Y_5 = -1 + 0.25j$

$Y_6 = 0.2269 - 1.7208j$

$Y_7 = 0.8047 + 0.3499j$

(inverse!

$y(n) = ?$